

Air Quality Index (AQI) Prediction using Machine Learning

Linear Regression Model with AQI & Location Data

Objective

- Analyze Global Air Quality Index (AQI)
- Build a predictive model
- Visualize AQI distribution
- Enable interactive AQI mapping

Goal of the project

- Develop a robust supervised regression model leveraging historical multi-parameter air quality datasets—including PM_{2.5}, NO₂, SO₂, CO, O₃, to predict future AQI values.
- Optimize model performance through feature engineering and validation techniques to ensure high predictive accuracy.
- Enable real-time forecasting to support environmental engineers and regulatory agencies in proactive emission control and public health risk mitigation

Approach



1

Data Collection:

- Source: GeeksforGeeks dataset
- Features: PM2.5, NO₂, CO, O₃, lat, & lng
- Relevant factors directly linked to AQI

2

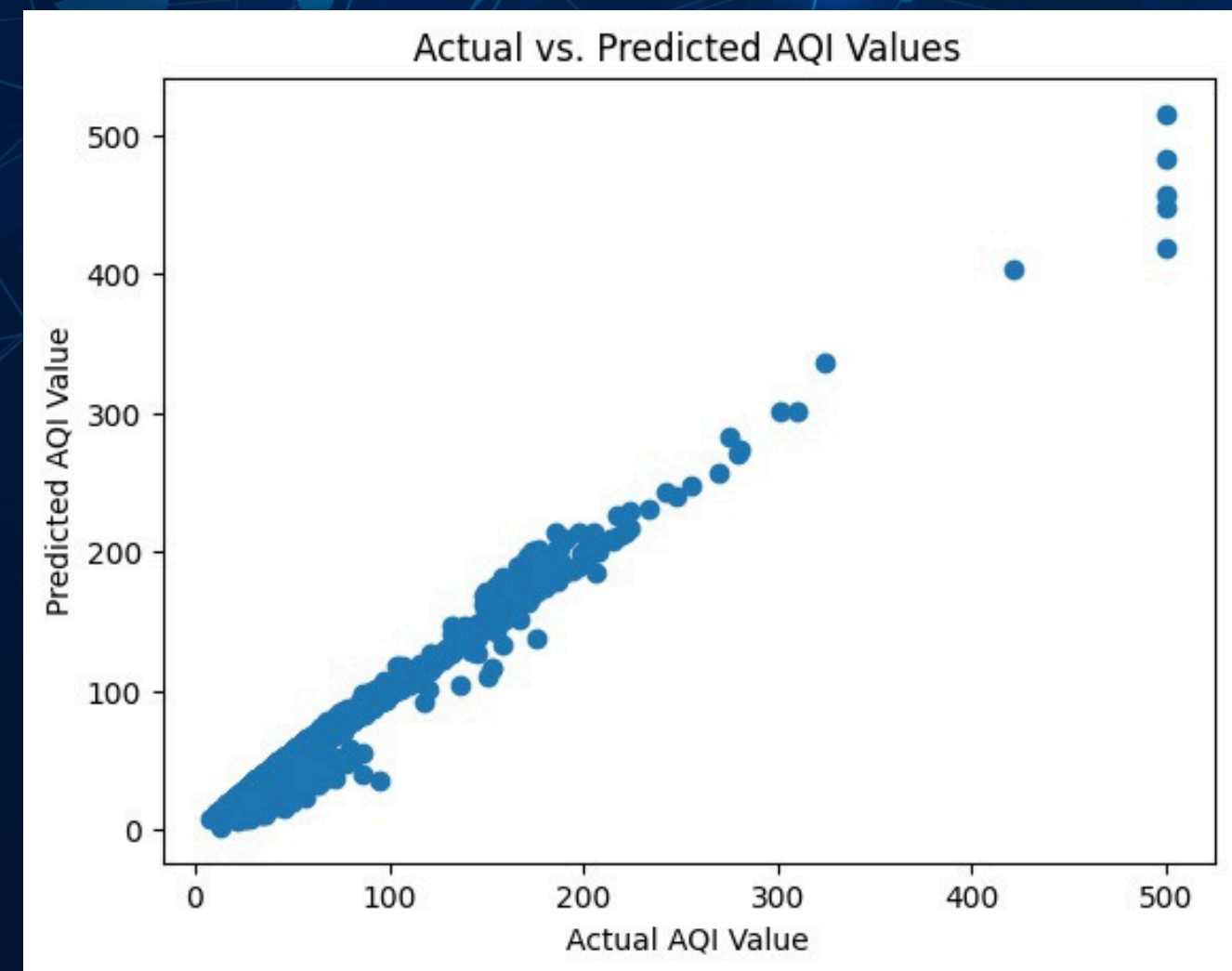
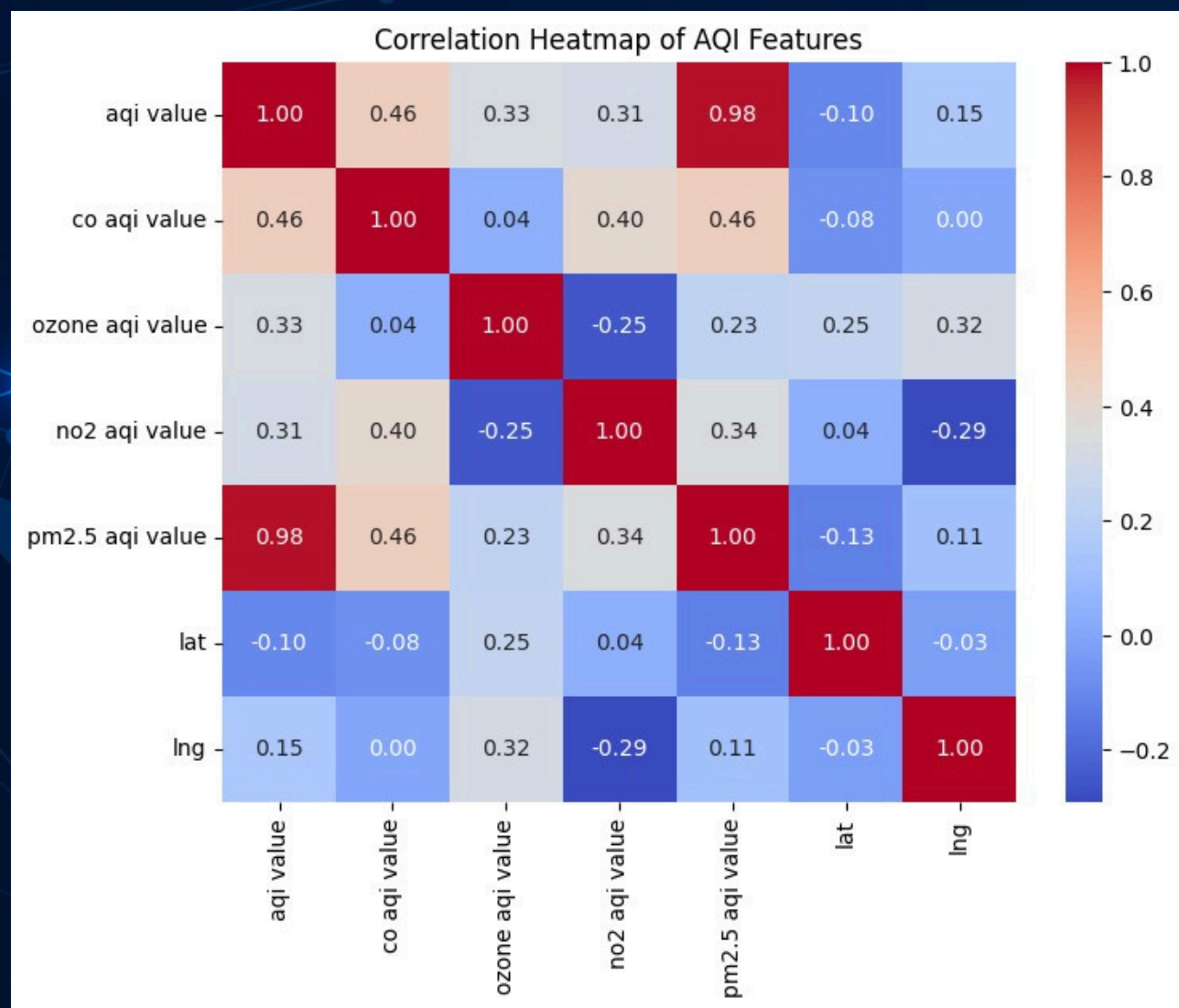
Data Pre-processing:

- Dataset already clean
- Performed exploratory analysis (heatmaps, pairplots)
- Split into features & target (AQI)
- Applied train-test split for fair evaluation

3

Model Development:

- Linear Regression chosen
 - Simple, interpretable, efficient
 - Shows contribution of each pollutant
- KMeans clustering for grouping AQI levels



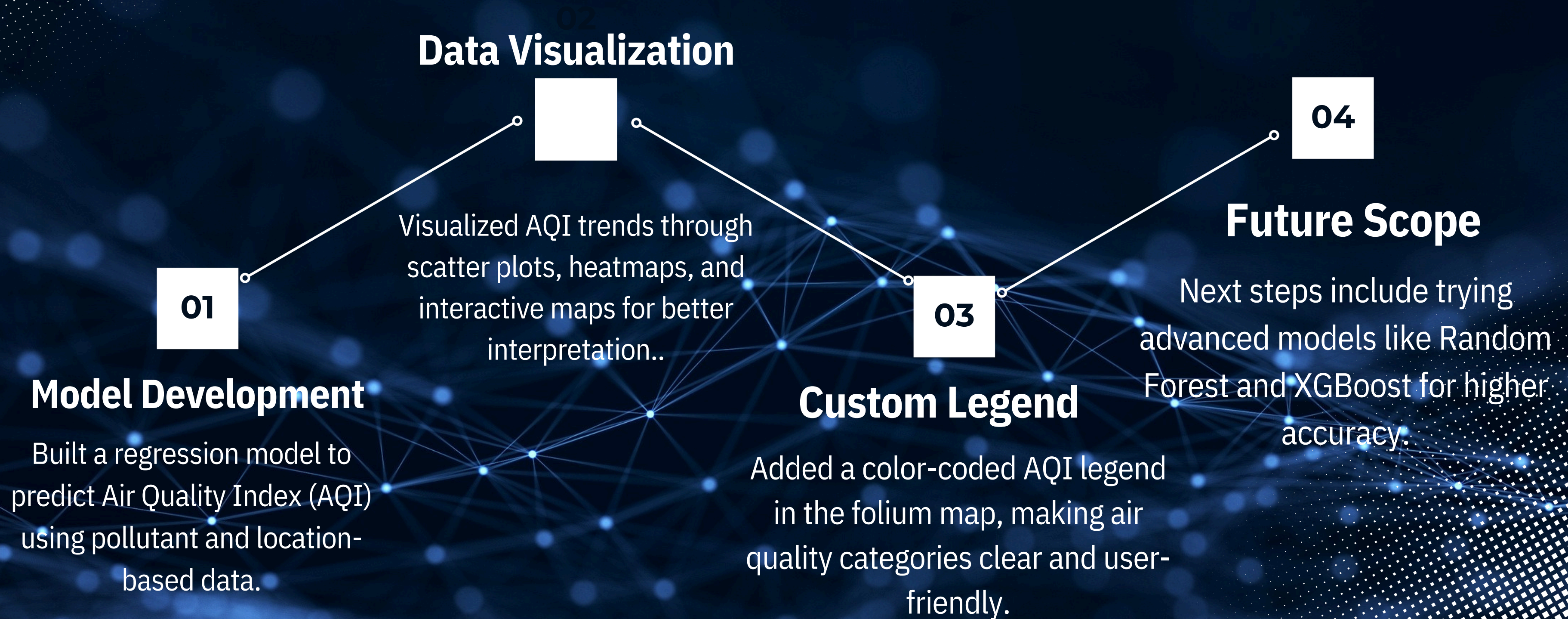
Performance Evaluation

- Mean Absolute Error (MAE): 4.17
- On average, predictions are only ~4 AQI units away from the actual values.
- R^2 Score: 0.974
- Model explains 97.4% of the variation in AQI values.
- Indicates high accuracy and low error rate.

Visualization with Folium

- Used Folium for interactive mapping.
- AQI values plotted on map with latitude & longitude.
- Hover feature: Instantly view AQI level of any location.
- Makes results easy to interpret and visually engaging.

CONCLUSION





THANK YOU!